

Oracle EPM Predictive Cash Forecasting

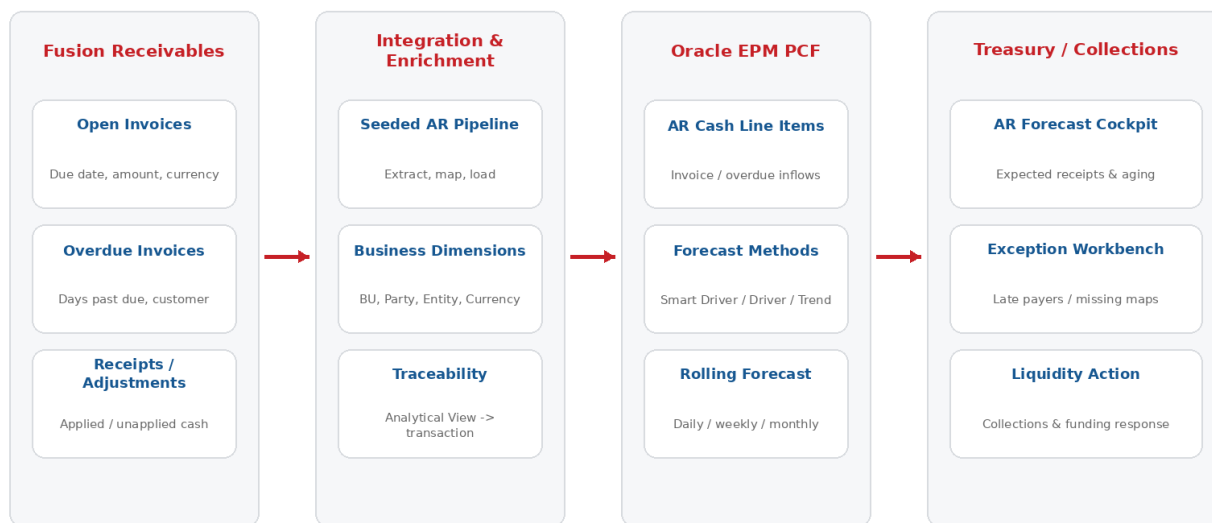
AR Forecasting

Advanced Lab Guide | Product Design + Implementation + Technical Reference + Troubleshooting

Advanced-stage lab objective

Design, configure, run, reconcile and troubleshoot a transaction-driven Accounts Receivable cash forecast using Oracle EPM Predictive Cash Forecasting, with Fusion Receivables integration and source traceability.

AR Forecasting Reference Architecture



Lab profile	Design choice
Forecast horizon	13 weeks (weekly) with short-term Smart Driver focus
Primary source	Oracle Fusion Cloud Receivables
Core line items	Receivable Invoice; Receivable Overdue Invoice
Scenario set	Base; Stress; Management Action
Primary controls	Mapping, due dates, aging, security, traceability, FVA

1. Purpose, Scope and Learning Outcomes

AR Forecasting converts open receivable transactions into expected customer cash receipts. In a strong implementation, the forecast is transaction-grounded, method-aware, explainable to source, and operationally linked to collections actions and liquidity decisions.

- Design the AR cash-forecast product model and user journey.
- Configure source, dimensions, line items and forecast methods.
- Use invoice due dates and overdue behavior to build a 13-week receipt forecast.
- Apply Base, Stress and Management Action scenarios.
- Drill through from PCF to Fusion Receivables analytical views and detailed transactions.
- Measure forecast accuracy and Forecast Value Add (FVA).
- Operate controls, pipeline jobs, security, audit and troubleshooting.

Current Oracle capability note

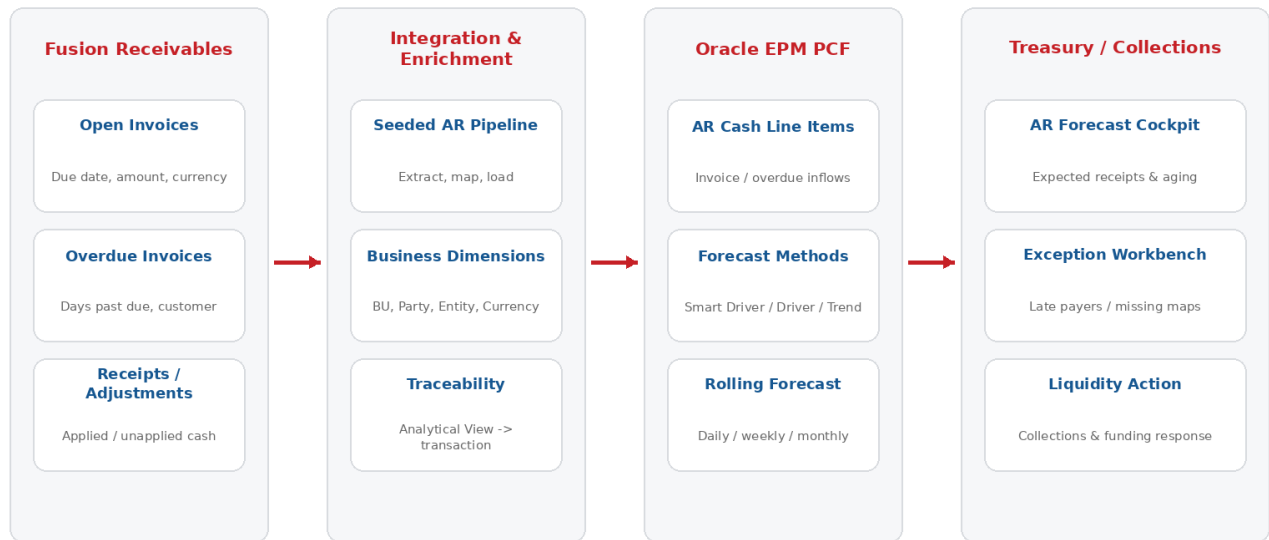
Oracle PCF integrates with Fusion Receivables, Payables and Cash Management through ready-to-use integrations and pre-seeded Data Integration pipelines. Drill-through can reach aggregate Analytical Views and detailed ERP transactions. Smart Drivers use near-real-time transactional data and are particularly valuable for short-term forecasting.

Audience

Role	Use of this guide
Oracle EPM consultant	Configuration, dimensions, methods, rules, jobs
Fusion Financials consultant	Receivables readiness, roles, data access, source traceability
Treasury / Cash manager	Receipt outlook, liquidity risk, scenario actions
Collections manager	Customer prioritization and overdue recovery
Support engineer	Job monitoring, reconciliation and incident recovery

2. AR Forecasting Functional Architecture

AR Forecasting Reference Architecture

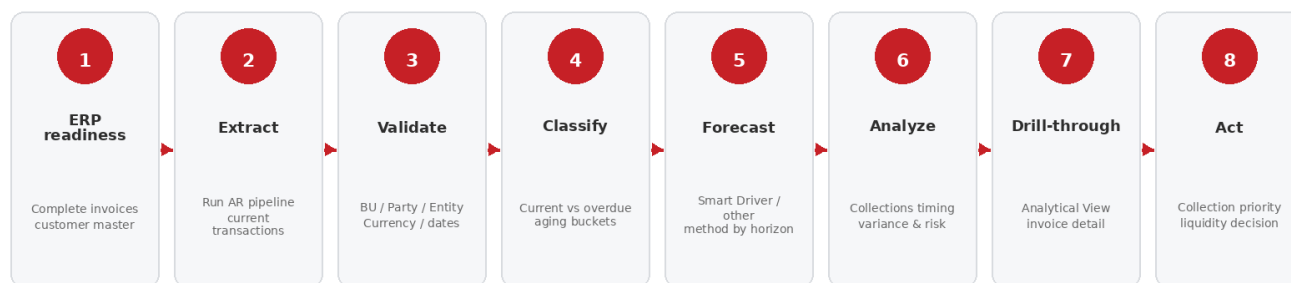


Design principles

- Forecast the direct cash receipt, not accounting revenue.
- Keep current invoices and overdue invoices distinguishable for behavior and controls.
- Preserve Party/Customer and Business Unit context for analysis and source traceability.
- Use transactional Smart Drivers for near-term horizons when data quality is operationally ready.
- Use driver/trend/statistical or manual methods where transaction data is not sufficient for the horizon.

3. End-to-End AR Forecasting Process

AR Forecasting End-to-End Process



Operational cadence

Stage	Owner	Control / evidence
ERP readiness	AR Operations	Invoices, installments, due dates, customer master current
Pipeline	EPM Admin	Successful extract/load job and record counts
Validation	Treasury Analyst	Unmapped Party/BU/Entity/Currency = 0 critical
Forecast	Cash Manager	Methods executed for selected horizon
Review	Collections + Treasury	Top overdue/at-risk customers reviewed
Approval	Treasury Manager	Scenario and action plan approved

4. Data Model and AR Line-Item Design

Core intersections

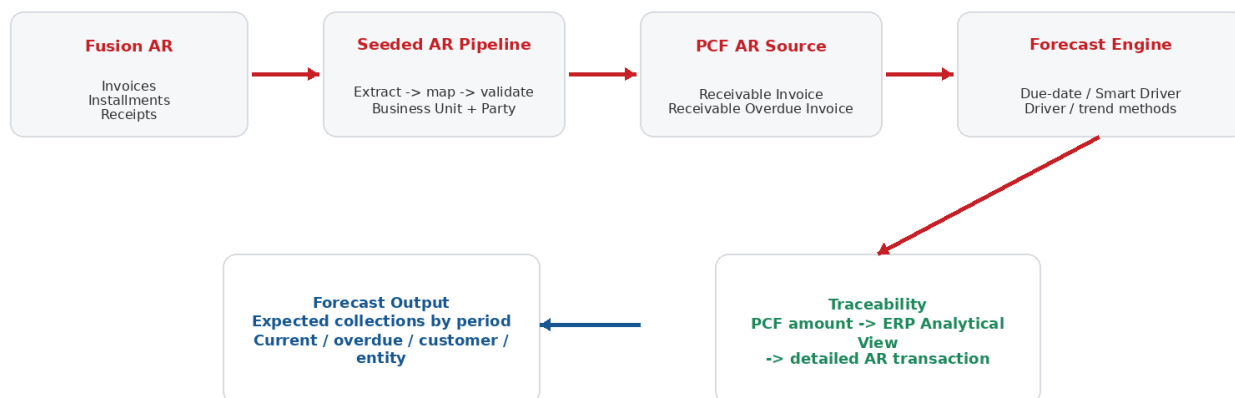
Dimension / attribute	AR purpose	Example
Cash Flow Line Item	Classifies forecasted inflow	Receivable Invoice
Forecast Method	Method by horizon	Smart Driver
Business Unit	Fusion operating context	US Operations BU
Party	Customer / counterparty	ACME-EU
Entity	Legal entity liquidity owner	US01
Currency	Transaction/reporting currency	USD
Period	Daily/weekly/monthly bucket	FY26-W36
Scenario	Forecast version	Base

Recommended AR cash line items

Line item	Meaning	Typical forecast treatment
Receivable Invoice	Open, not overdue invoice installments	Due-date / Smart Driver
Receivable Overdue Invoice	Past-due invoice installments	Behavior/overdue assumptions + review
Other Customer Receipts	Non-invoice inflows if modeled	Driver/manual/source-specific
AR Adjustment / Override	Approved treasury adjustment	Controlled manual override

5. Source-to-Forecast Data Flow and Traceability

AR Data Flow, Forecast Logic and Drill-Through



Control principle: every AR forecast amount should retain source period, customer/party, business unit, entity, currency, forecast method and t

Traceability requirement

For every material forecast amount, users should be able to explain which source data, mapping, period, entity, customer/party, currency and forecast method contributed to it. Where Fusion ERP integration is enabled, drill-through provides a strong investigation path.

6. Fusion Receivables Integration Design

Readiness checklist

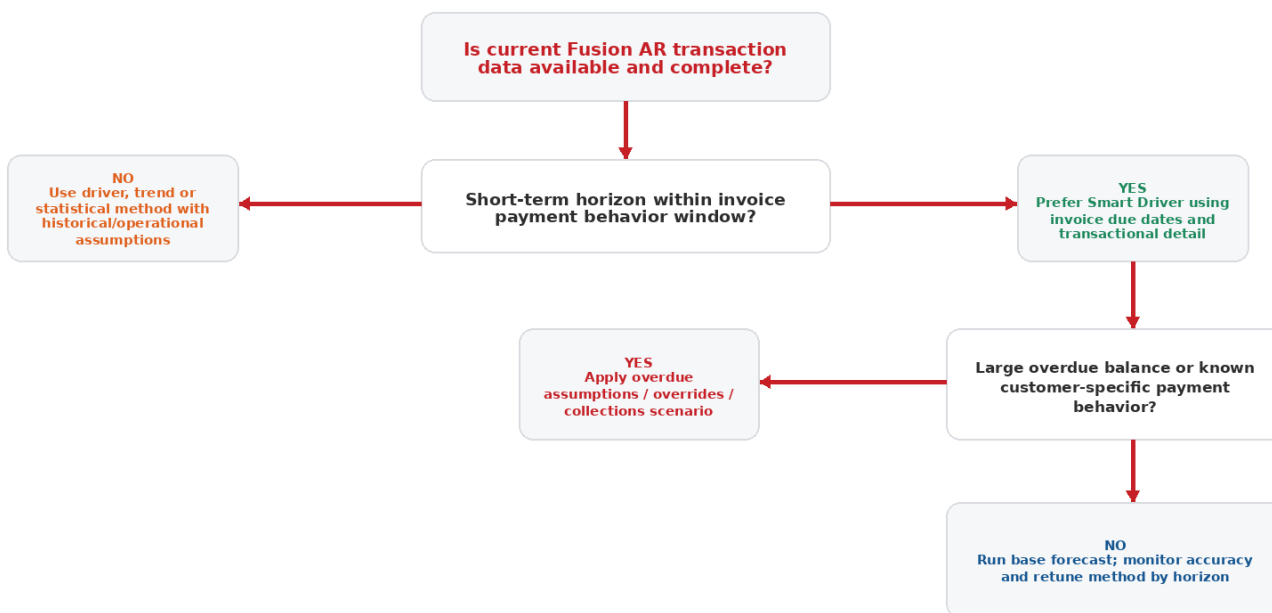
- Enable the PCF integration prerequisites in Fusion ERP and EPM.
- Ensure Analytical Views are enabled where drill-through is required.
- Provision users in both Fusion ERP and Cloud EPM, with aligned entity/data access.
- Enable Data Sources during initial PCF feature enablement when Fusion integration is required.
- Validate Business Unit and Party dimensions and mappings before production load.
- Keep AR transaction processing current; stale invoices or receipts directly degrade forecast quality.

Integration control matrix

Control	Test	Failure symptom
Connectivity	Seeded AR pipeline connects	No extract / authentication error
Source completeness	Invoice/receipt counts vs ERP	Understated forecast
Party mapping	All material customers mapped	Rejects / Other bucket
BU security	Integration user has BU access	Partial extract / access denied
Date quality	Due date/installment date valid	Receipts shifted to wrong week
Drill-through	PCF -> Analytical View -> transaction	Traceability unavailable

7. Forecast Method Decision Framework

AR Forecast Method Decision Tree



Method guidance

Situation	Preferred approach	Why
Near-term current invoices	Smart Driver	Uses detailed ERP invoice due dates
Near-term overdue invoices	Smart Driver + overdue assumptions / review	Past due behavior needs explicit treatment
Longer horizon without invoices	Driver / trend / statistical	Future invoices do not yet exist
Known one-off receipt	Manual controlled override	Business event may not be inferable
Stable historical seasonal pattern	Trend / Predictive Planning	Historical behavior can add value

ML caution

Oracle documentation currently describes ML forecast methods but notes machine learning support as a future update. Do not design production dependency on ML until it is available in the tenant/release you operate.

8. Oracle-Inspired AR Configuration Mockup

Oracle-inspired | Predictive Cash Forecasting

AR Forecasting Setup

AR Forecast Configuration

Data Integration

Forecast Methods

Cash Line Items

Smart Drivers

Scenario

Security

Jobs

Receivable Invoice - Forecast Method

Method	Forecast Range	Primary Date	Source	Drill-Through
Smart Driver	W1-W8	Invoice Installment Due Date	Fusion Cloud Receivables	Enabled

Overdue Invoice Assumptions

1-30 days	70% next 2 weeks
31-60 days	50% next 4 weeks
61-90 days	30% next 6 weeks
90+ days	Manual / collections review

Critical Controls

- Business Unit mapped
- Party/customer mapped
- Entity security aligned
- Currency valid
- Due date present
- Duplicate invoice blocked
- Source timestamp within SLA

Configuration checkpoints

- Assign forecast methods by line item and forecast range.
- Use installment due date as the minimum transactional timing anchor for AR Smart Drivers.
- Define explicit overdue behavior for 1-30, 31-60, 61-90 and 90+ day buckets.
- Restrict manual overrides and require commentary/approval for material values.
- Align source data access and entity security between ERP and EPM.

9. Advanced Lab Scenario - 13-Week AR Forecast

BISP Global Services has USD 14.8M of open receivables. Treasury needs a 13-week cash-in forecast. The first eight weeks use transaction-driven logic; later weeks use driver assumptions because future invoices are not yet available.

Parameter	Value
Opening cash position	\$6.0M
Open AR	\$14.8M
Overdue AR	\$4.1M
Forecast horizon	13 weeks
Minimum cash buffer	\$4.5M
Base collection behavior	Current invoices: due-date based; overdue: aging assumptions
Stress assumption	20% delay on top 10 customers + 10% overall timing slip

AR aging at lab start

Bucket	Balance	Base receipt assumption
Current / not due	\$10.7M	Due date / Smart Driver
1-30 days overdue	\$2.0M	70% in next 2 weeks
31-60 days overdue	\$1.0M	50% in next 4 weeks
61-90 days overdue	\$0.7M	30% in next 6 weeks
90+ days overdue	\$0.4M	Manual collections review

10. Lab Step 1 - Build the Base AR Receipt Forecast

Week	Expected AR cash receipt	Cumulative	Primary method
W1	\$2.10M	\$2.10M	Smart Driver
W2	\$2.40M	\$4.50M	Smart Driver
W3	\$2.00M	\$6.50M	Smart Driver
W4	\$1.70M	\$8.20M	Smart Driver
W5	\$1.90M	\$10.10M	Smart Driver
W6	\$2.20M	\$12.30M	Smart Driver
W7	\$2.00M	\$14.30M	Smart Driver
W8	\$1.80M	\$16.10M	Smart Driver
W9	\$2.10M	\$18.20M	Driver / trend
W10	\$2.30M	\$20.50M	Driver / trend
W11	\$2.00M	\$22.50M	Driver / trend
W12	\$2.20M	\$24.70M	Driver / trend
W13	\$2.40M	\$27.10M	Driver / trend

Calculation pattern

Expected AR Cash Receipt(period) = current invoice installments expected in period + overdue recovery expected in period + approved AR overrides. The exact component logic depends on the selected PCF method and source configuration.

Control test

For W1-W4, reconcile forecast receipts to invoice-level or analytical-view evidence. Investigate differences caused by overdue routing, disputed invoices, missing Party mappings, currency conversion or timing assumptions.

11. Lab Step 2 - Customer-Level Collection Risk

Customer	Open AR	Overdue	Base next-4W receipt	Risk signal	Action
ACME-EU	\$1.20M	\$0.42M	\$0.72M	18 days late	Collector escalation
Nova Retail	\$0.95M	\$0.31M	\$0.46M	Dispute	\$200K resolution plan
Zenith Inc	\$0.80M	\$0.26M	\$0.39M	Promise broken	Executive follow-up
Orion GmbH	\$0.62M	\$0.18M	\$0.34M	Mapping review	Fix Party mapping
Other	\$11.23M	\$2.93M	\$4.82M	Portfolio	Normal cadence

Advanced exercise

- Rank customers by cash-at-risk = overdue balance x probability of delay.
- Compare concentration risk: top 10 customer forecast receipts as % of total next 4 weeks.
- Create a collections action scenario that improves only the overdue component, not current-invoice due dates.

12. Lab Step 3 - Stress Scenario and Management Action

Week	Base	Stress	Mgmt Action	Stress gap
W1	\$2.10M	\$1.80M	\$1.95M	\$0.30M
W2	\$2.40M	\$1.90M	\$2.10M	\$0.50M
W3	\$2.00M	\$1.70M	\$1.85M	\$0.30M
W4	\$1.70M	\$1.50M	\$1.60M	\$0.20M
W5	\$1.90M	\$1.60M	\$1.75M	\$0.30M
W6	\$2.20M	\$1.90M	\$2.00M	\$0.30M
W7	\$2.00M	\$1.70M	\$1.85M	\$0.30M
W8	\$1.80M	\$1.60M	\$1.70M	\$0.20M
W9	\$2.10M	\$1.80M	\$1.95M	\$0.30M
W10	\$2.30M	\$1.90M	\$2.05M	\$0.40M
W11	\$2.00M	\$1.80M	\$1.90M	\$0.20M
W12	\$2.20M	\$1.90M	\$2.05M	\$0.30M
W13	\$2.40M	\$2.00M	\$2.15M	\$0.40M

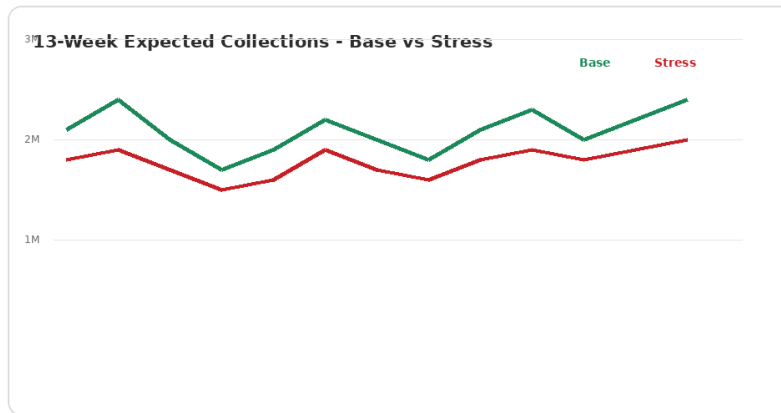
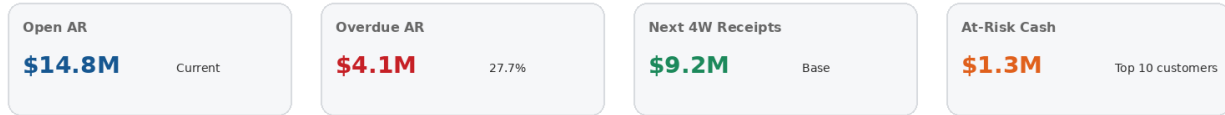
Scenario interpretation

Stress delays collections, which can create a liquidity gap even when accounting receivables remain unchanged. Management Action should represent specific collection interventions, dispute resolution, payment-plan acceleration or treasury funding action - not an unexplained optimistic override.

13. AR Forecast Cockpit / Dashboard Mockup

Oracle-inspired | AR Forecast Cockpit

Accounts Receivable Cash Forecast



Collection Exceptions

ACME-EU \$420K	18d late
Nova Retail \$315K	Dispute
Zenith Inc \$260K	Promise broken
Orion GmbH \$185K	Mapping review

Recommended KPIs

KPI	Purpose
Open AR	Total receivable exposure
Overdue AR %	Portfolio collection risk
Next 4W expected receipts	Near-term cash visibility
At-risk cash	Material delay exposure
Top customer concentration	Dependency on large payers
Forecast vs Actual	Model reliability
Promise-to-pay hit rate	Collections execution quality

14. Forecast vs Actual and Forecast Accuracy

Week	Forecast	Actual	Variance	Abs % Error
W1	\$2.10M	\$2.00M	\$-0.10M	5.0%
W2	\$2.40M	\$2.25M	\$-0.15M	6.7%
W3	\$2.00M	\$1.72M	\$-0.28M	16.3%
W4	\$1.70M	\$1.85M	\$+0.15M	8.1%
W5	\$1.90M	\$1.66M	\$-0.24M	14.5%
W6	\$2.20M	\$2.05M	\$-0.15M	7.3%

Use variance attribution, not only a single accuracy score. Separate timing variance, amount variance, invoice dispute, credit memo/adjustment, late payment, mapping defects and manual override impact.

Forecast Value Add (FVA)

Compare each enhanced method against a simple baseline. If a complex adjustment consistently worsens accuracy, remove or redesign it. FVA should be monitored by line item, horizon, entity and customer segment where meaningful.

15. Overdue AR Forecasting and Collections Integration

Aging bucket	Forecast risk	Recommended control
1-30 days	Moderate timing slippage	Promise-to-pay / collector review
31-60 days	Higher timing uncertainty	Customer-specific probability / scenario
61-90 days	Material recovery risk	Credit + collections escalation
90+ days	Low predictive reliability	Manual review, dispute/legal status, conservative scenario

Design principle

Do not let heavily overdue balances flow automatically through the same logic as current invoices without explicit assumptions. They often require customer-specific judgment, dispute status, promise-to-pay evidence or conservative collection scenarios.

16. Drill-Through and Transaction Traceability

- Start from a PCF forecast or actual amount for AR.
- Drill to the Fusion Receivables aggregate Analytical View where configured.
- Filter by customer, Business Unit, invoice status, due date or aging characteristics.
- Continue to detailed transaction view to inspect invoice/installment details.
- Use source evidence to explain variance, customer delay and forecast confidence.

Security dependency

Users must be appropriately provisioned in both Cloud ERP and Cloud EPM. Entity/data access should be aligned, and SSO/enterprise connections should support the cross-application drill-through experience.

17. Pipeline Automation and Job Orchestration

Sequence	Job / gate	Pass criteria
1	Validate ERP connection	Connection successful
2	Run seeded AR Data Integration pipeline	Status Success / accepted exceptions
3	Validate record counts and rejects	Critical rejects = 0
4	Run mappings / business rules	No unmapped material records
5	Execute AR forecast methods	All selected ranges complete
6	Run aggregate / data maps if required	Target refreshed
7	Publish dashboard / variance	Refresh successful
8	Notify reviewers	Workflow task issued

Restart strategy

- Design idempotent loads where possible.
- Checkpoint after source load and after forecast calculation.
- Do not rerun successful downstream steps blindly after a source correction.
- Persist run ID, period, source timestamp, record counts and rejected-record evidence.

18. Security, Governance and Audit

Role	Typical responsibility	Restriction
Service Administrator	Application/integration admin	No business override without governance
Treasury Manager	Approve forecast/scenarios	Entity scope
Treasury Analyst	Analyze and prepare	No final approval
Collections Manager	Customer collection actions	AR scope
Integration Operator	Run/restart pipelines	No forecast approval
Audit / Read-only	Evidence review	No write access

- Require comments for material overrides.
- Capture before/after value, user, timestamp, reason and workflow status.
- Segregate integration operation from forecast approval.
- Align EPM entity access with Fusion ERP data access for traceability.

19. Log and Error Examples

AR Integration / Forecast Error Console

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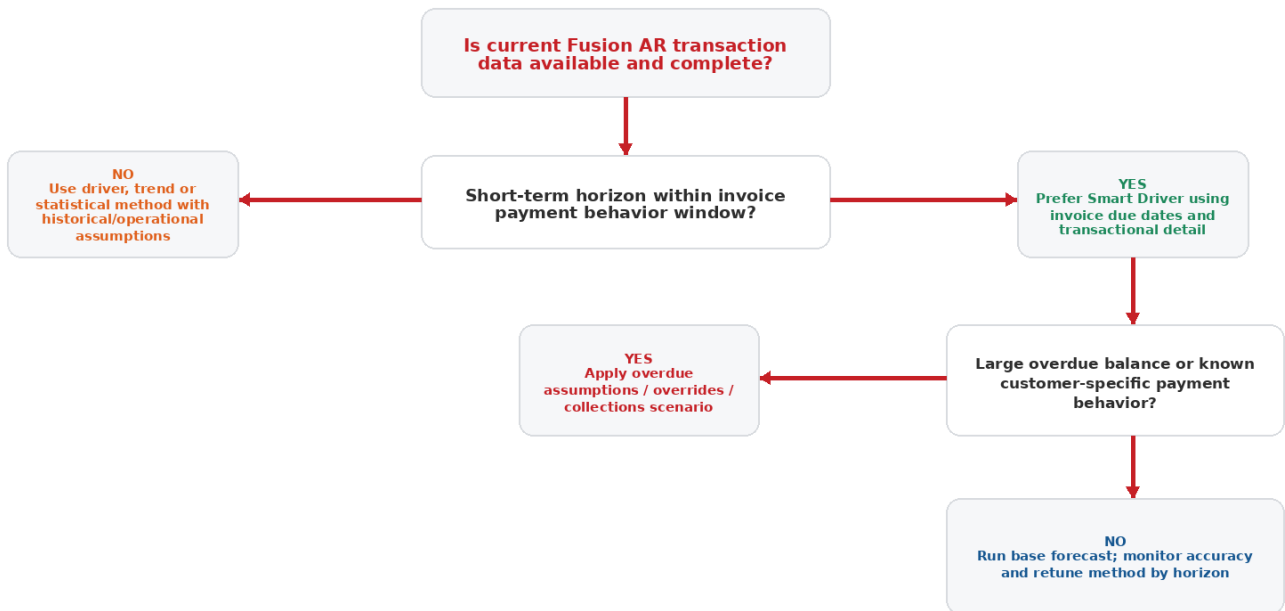
07:45:02 INFO    PCF_AR_PIPELINE started | Forecast=W35-W47
07:45:21 INFO    Extracted 48,326 invoice installments
07:45:29 WARN    312 invoices missing promised receipt date; due date used
07:45:41 ERROR   PARTY_9182 has no target Party mapping | Customer=ACME-EU
07:45:55 ERROR   BU_US03 not accessible to integration user
07:46:14 WARN    1,204 invoices >90 days overdue routed to manual review
07:46:40 INFO    Loaded Receivable Invoice=39,220 | Overdue=8,782
07:47:12 ERROR   Drill-through privilege missing for Receivables Analytical View
07:47:28 INFO    Job status=PARTIAL_SUCCESS | Reject=324
  
```

Error classification

Error class	Examples	Response
Source readiness	Stale invoice/receipt data	Fix ERP operational process, rerun extract
Mapping	Party/BU/Entity unmapped	Correct mapping and reprocess rejected records
Security	BU or Analytical View denied	Correct roles/data access/SSO
Data quality	Missing due date / invalid currency	Repair source or controlled transformation
Forecast logic	Wrong method/range	Correct method assignment and recalculate
Performance	Long extract/load	Review volume, filters, cadence, job concurrency

20. Troubleshooting Decision Tree

AR Forecast Method Decision Tree



Fast diagnostic path

- First determine whether the problem is source data, integration, mapping, forecast method, security, or aggregation.
- Compare ERP source counts/amounts with EPM loaded counts/amounts before changing forecast logic.
- For missing values, inspect Party and Business Unit mappings before assuming calculation failure.
- For strange timing, verify installment due dates and overdue classification.
- For drill-through failure, validate Analytical Views, roles, data access, SSO and enterprise connection.

21. Production Runbook

Time / cadence	Activity	Owner	Evidence
Daily / before forecast	Confirm Fusion AR close-to-current	AR Ops	Source freshness timestamp
Forecast cycle	Run AR pipeline	Integration Ops	Job ID + counts
Post-load	Review rejects / mappings	EPM Admin	Exception report
Calculation	Run AR forecast	Treasury Analyst	Forecast job result
Review	Analyze top customer risks	Collections/Treasury	Commentary/action log
Approval	Approve Base / Stress	Treasury Manager	Workflow approval
Post-period	Forecast vs Actual / FVA	Treasury COE	Accuracy dashboard

22. Implementation Checklist

- ☐ Confirm EPM Enterprise subscription and separate PCF application instance requirements.
- ☐ Enable PCF features and Data Sources correctly during initial enablement.
- ☐ Configure Fusion ERP prerequisites and Analytical Views if drill-through is required.
- ☐ Validate SSO and user provisioning in both applications.
- ☐ Define Business Unit, Party, Entity, Currency and line-item mappings.
- ☐ Load representative AR history and current transactions.
- ☐ Configure current and overdue AR forecast methods by horizon.
- ☐ Design Base, Stress and Management Action scenarios.
- ☐ Test drill-through to aggregate and detailed Receivables data.
- ☐ Establish record-count, amount and reject controls.
- ☐ Configure security, override governance and approval workflow.
- ☐ Automate jobs with restart/checkpoint rules.
- ☐ Run forecast-vs-actual and FVA during parallel testing.
- ☐ Complete performance and volume testing.
- ☐ Document support runbook and ownership.

23. Advanced Student Challenge

Using the lab dataset, redesign the AR forecast for a multinational business with USD, EUR and GBP receivables, three Business Units and customer concentration risk. Your submission should include:

- Source-to-target mapping matrix.
- Forecast method by line item and horizon.
- Overdue aging assumptions by customer segment.
- 13-week Base, Stress and Management Action forecast.
- Liquidity impact and collection action plan.
- Forecast-vs-actual/FVA framework.
- Security and drill-through design.
- Pipeline restart and incident response plan.

Assessment standard

A strong answer is traceable, controlled and operational. It should explain not only the forecast number, but why cash is expected in that period, which invoices/customers drive it, what could make it late, and what action treasury or collections should take.

24. Technical Reference and Key Oracle Notes

Topic	Reference point
AR source	Oracle Fusion Cloud Receivables
Seeded forecast line items	Receivable Invoice; Receivable Overdue Invoice
Smart Driver basis	Transactional data; due date of invoice installment at minimum
Forecast frequencies	Daily, weekly or monthly rolling forecasts
Traceability	PCF -> ERP aggregate Analytical View -> detailed transaction
Core integration dimensions	Business Unit and Party are important seeded source dimensions
Platform	Predictive Cash Forecasting is a Cloud EPM Planning application type
Currency	PCF applications are multi-currency

Oracle documentation also recommends operationally current and complete ERP data for reliable forecasts. Historical Receivables transaction history is valuable for model analysis; earlier integration guidance recommended at least 18 months of complete AR transactional activity to maximize model-training potential where applicable.

Release-awareness note

Capabilities evolve by Cloud EPM release. Validate forecast methods, integration features and any ML-related functionality in the target tenant before committing them to a production solution design.